

APOTI DEVELOPMENT ASSOCIATION

Borehole Drilling and Solar Pump Installation — Nkwen-Mbatu

Financial Report

ADA/2024/002 · June 1 – October 31, 2024

FINANCIAL source markdown

2,700,000

Prepared for public download

REFERENCE
ADA/2024/002

LOCATION
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PERIOD
June 1 – October 31, 2024

BUDGET / SPEND
2,700,000

Financial Report

Clean Water Well — Nkwen-Mbatu Village

FIELD	DETAIL
Reference	ADA/2024/002
Project Title	Borehole Drilling and Solar Pump Installation — Nkwen-Mbatu
Implementing Organization	Apoti Development Association (ADA)
Reporting Period	June 1 – October 31, 2024
Currency	CFA Franc (XAF)
Exchange Rate	1 USD = 600 CFA (approximate)

1. Budget vs. Actual Summary

#	BUDGET LINE	BUDGETED (CFA)	ACTUAL (CFA)	VARIANCE (CFA)	% SPENT
1	Geological survey	150,000	150,000	0	100%
2	Borehole drilling (45 m)	1,500,000	1,450,000	-50,000	97%
3	Solar pump + installation	600,000	600,000	0	100%
4	Piping and fittings	200,000	200,000	0	100%
5	Water committee training	50,000	50,000	0	100%
6	Community mobilization/meetings	50,000	50,000	0	100%
7	Transport/coordination	80,000	80,000	0	100%
8	Documentation/signage	30,000	30,000	0	100%
9	Contingency	40,000	40,000	0	100%
	TOTAL	2,700,000	2,650,000	-50,000	98%

USD Equivalent

	CFA	USD
Total Budget	2,700,000	\$4,500
Total Actual	2,650,000	\$4,417
Savings	50,000	\$83

2. Variance Explanation

Line 2: Borehole Drilling (-50,000 CFA)

The drilling contractor's quote was based on a target depth of 45 meters at a rate of approximately 33,333 CFA per meter. Water was struck at 42 meters depth — 3 meters shallower than estimated — due to favorable geological conditions (fractured zone

encountered earlier than predicted by the survey). The contractor invoiced for actual depth drilled (42 m), resulting in a saving of 50,000 CFA.

Line 9: Contingency (0 CFA variance, fully utilized)

The contingency fund of 40,000 CFA was allocated as follows:

- Additional community meetings (unplanned 4th meeting due to high interest): 15,000 CFA
- Road repair materials for drilling rig access after heavy rains: 15,000 CFA
- Additional protective packaging for solar panel transport: 10,000 CFA

Savings Allocation

The 50,000 CFA savings from reduced drilling depth was transferred to the village water committee's maintenance fund as a startup contribution, per agreement between ADA and the committee.

3. Detailed Expenditure by Category

3.1 Geological Survey — 150,000 CFA

ITEM	AMOUNT (CFA)
Hydrogeologist professional fee (2 days)	100,000
VES equipment rental	30,000
Transport to site	15,000
Report preparation	5,000
Subtotal	150,000

3.2 Borehole Drilling — 1,450,000 CFA

ITEM	AMOUNT (CFA)
Drilling rig mobilization/demobilization	200,000
Drilling (42 m × rate)	840,000
PVC casing (42 m, 6-inch)	170,000
Stainless steel screen (7 m)	90,000
Gravel pack material	40,000
Cement grout (sanitary seal)	30,000
Well development (6 hours)	50,000
Pump test (8 hours)	30,000
Subtotal	1,450,000

3.3 Solar Pump + Installation — 600,000 CFA

ITEM	AMOUNT (CFA)
Submersible solar pump	250,000
Solar panels (4 × 100W)	200,000
MPPT controller + wiring	50,000

Mounting frame (galvanized steel)	40,000
Elevated tank (3,000 L polyethylene)	35,000
Tank tower (steel/concrete, 4 m)	15,000
Installation labor	10,000
Subtotal	600,000

3.4 Piping and Fittings — 200,000 CFA

ITEM	AMOUNT (CFA)
HDPE rising main (pump to surface)	45,000
PVC distribution piping (tank to tap stands)	55,000
Brass taps and fittings (2 tap stands)	30,000
Concrete aprons (2 units)	40,000
Drainage channels	15,000
Wellhead cover (lockable steel)	10,000
Perimeter fencing materials	5,000
Subtotal	200,000

3.5 Water Committee Training — 50,000 CFA

ITEM	AMOUNT (CFA)
Facilitator fee (2 days)	20,000
Training materials and printing	10,000
Refreshments (5 participants × 2 days)	10,000
Maintenance manuals (5 copies, illustrated)	10,000
Subtotal	50,000

3.6 Community Mobilization — 50,000 CFA

ITEM	AMOUNT (CFA)
Community meetings (3 sessions)	20,000
Hygiene awareness sessions (3 sessions)	15,000
Public address system rental	10,000
Refreshments for meetings	5,000
Subtotal	50,000

3.7 Transport/Coordination — 80,000 CFA

ITEM	AMOUNT (CFA)
ADA coordinator transport (monthly visits × 5 months)	50,000
Fuel for supervision trips	20,000

Communication (phone credit)	10,000
Subtotal	80,000

3.8 Documentation/Signage — 30,000 CFA

ITEM	AMOUNT (CFA)
Project information board (fabrication + installation)	15,000
Borehole technical data plate	5,000
Photography/documentation	5,000
Printed project file for committee	5,000
Subtotal	30,000

3.9 Contingency (Utilized) — 40,000 CFA

ITEM	AMOUNT (CFA)
Additional community meeting (4th session)	15,000
Road repair materials for rig access	15,000
Solar panel protective packaging	10,000
Subtotal	40,000

4. Expenditure by Phase

PHASE	PERIOD	AMOUNT (CFA)	% OF TOTAL
Phase 1: Preparation	June 2024	250,000	9%
Phase 2: Drilling	July–August 2024	1,530,000	58%
Phase 3: Solar Installation	September 2024	800,000	30%
Phase 4: Training & Handover	October 2024	70,000	3%
Total		2,650,000	100%

5. Cost Efficiency Analysis

METRIC	VALUE
Cost per beneficiary	~8,833 CFA (\$14.72 USD) per person
Cost per meter drilled	~34,524 CFA (\$57.54 USD) per meter
Daily water output	4,000–5,000 liters
Cost per liter/day capacity	~530–663 CFA (\$0.88–1.10 USD)
Recurring operational cost	0 CFA (solar-powered)
Estimated system lifespan	15–20 years
Cost per beneficiary per year (over 15 years)	~589 CFA (\$0.98 USD)

6. Payment Schedule

#	DATE	PAYEE	DESCRIPTION	AMOUNT (CFA)
1	June 5, 2024	Hydrogeologist	Geological survey (full payment)	150,000
2	June 15, 2024	Various	Community mobilization expenses	50,000
3	July 10, 2024	Drilling Contractor	Mobilization advance (50%)	725,000
4	Aug 5, 2024	Drilling Contractor	Completion balance	725,000
5	Sept 1, 2024	Solar Installer	Equipment + installation	600,000
6	Sept 15, 2024	Plumber/Supplier	Piping and fittings	200,000
7	Oct 1, 2024	Various	Training, transport, documentation	160,000
8	Various	Various	Contingency items	40,000
			Total Disbursed	2,650,000

7. Audit Notes

- All expenditures supported by receipts or invoices (filed in project folder)
- Drilling contractor invoice adjusted to reflect actual depth (42 m vs. quoted 45 m)
- Contingency expenditures pre-approved by ADA project lead
- 50,000 CFA savings formally transferred to water committee maintenance account
- No unauthorized expenditures or budget overruns recorded

8. Financial Conclusion

The project was delivered **50,000 CFA (1.9%) under budget**. All budget lines were managed within approved limits. The single variance (drilling depth) resulted in a saving that was productively allocated to the community's maintenance fund, strengthening long-term sustainability.

At approximately \$14.72 per beneficiary for a system with 15–20 year lifespan and zero recurring energy costs, this project represents exceptional value for money in the WASH sector.

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